Matt Eisenhauer

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EDUCATION

Georgia Institute of Technology | MS, Computational Data Analytics (GPA: 3.85) | Expected May 2027

Syracuse University | BS, Information Management & Technology (GPA: 3.41) | May 2018

SKILLS

Data Science & ML: Python (Pandas, NumPy, Scikit-Learn, PyTorch), R, Predictive Modeling, Ensemble Methods, Regularization, Unsupervised Learning

Statistical Analysis: Stochastic Modeling, Discrete-Event Simulation, Time-Series Forecasting, Experimental Design

Data Engineering & Visualization: SQL (Azure SQL / MS SQL Server), ETL/ELT, SQLAlchemy, Relational Database Design, Git, Matplotlib, Seaborn, Plotly, ggplot2

PROJECTS

Should I Stay or Should I Go? A Stochastic Simulation for Stolen Base Decision-Making

- Engineered a discrete-event simulation pipeline in Python that models stolen base attempts over 10,000 trials per matchup
- Ingested 2025 MLB Statcast data to create normal distributions for lead distance, pitcher delivery, and pop times
- Integrated an RE24 matrix framework to establish situational break-even success thresholds for "Go/No-Go" decisions
- Leveraged OpenCV for frame-by-frame video verification of raw game footage to validate pitcher time-to-plate distributions and backtest model accuracy

Shorten Up or Swing Away? A Machine-Learning Approach to Two-Strike Hitting

- Processed millions of rows of MLB Statcast data via pybaseball API, manipulating high-dimensional swing profile features
- Engineered a K-Means pipeline to cluster and identify distinct hitter archetypes based on two-strike swing profile delta
- Trained supervised regression models to predict xwOBA and evaluate the offensive impact of shortening up versus maintaining a traditional swing approach in two-strike counts.

Predictive Modeling of Elite Biomechanical Performance (Driveline Open Biomechanics Project)

- Preprocessed 1,900+ Driveline athlete profiles, deploying a 5-NN imputation framework to resolve missing tracking data
- Engineered custom anatomical asymmetry metrics to evaluate physical range-of-motion variances and rotational efficiency
- Trained regularized and ensemble models to predict peak pitch velocity and bat speed, identifying lower-body power metrics as the primary performance driver

WORK EXPERIENCE

UC Santa Barbara Baseball — Remote

Database Architect / Analytical Engineer (Freelance) — *June 2024 - Present*

- Engineered an automated ETL pipeline to ingest and process proprietary Trackman data across NCAA Division 1
- Designed and deployed an Azure SQL database of normalized play-by-play data optimized for predictive modeling
- Provided support for data quality issues by implementing preprocessing validation scripts to proactively flag errors
- Optimized relational schemas and indexing, slashing query latency and analyst processing times by hours

Bank of America — New York, NY

Vice President – Infrastructure Engineer / Architect — *January 2024 – Present*

- Designed scalable infrastructure and networking architectures for high-volume global banking applications, ensuring system resilience and seamless data flow across enterprise platforms
- Led incident response for core banking platforms, mobilizing cross-functional technical teams to accelerate recovery
- Awarded a US Patent for co-architecting an intelligent load-balancing solution that optimized dynamic traffic distribution and automated application failover across geographically distributed data centers

Consultant / AVP - Infrastructure Engineer — *July 2018 – December 2023*

- Developed Python automation scripts to streamline infrastructure design workflows, significantly reducing manual efforts
- Created interactive data visualizations and dashboards to monitor application health and security events